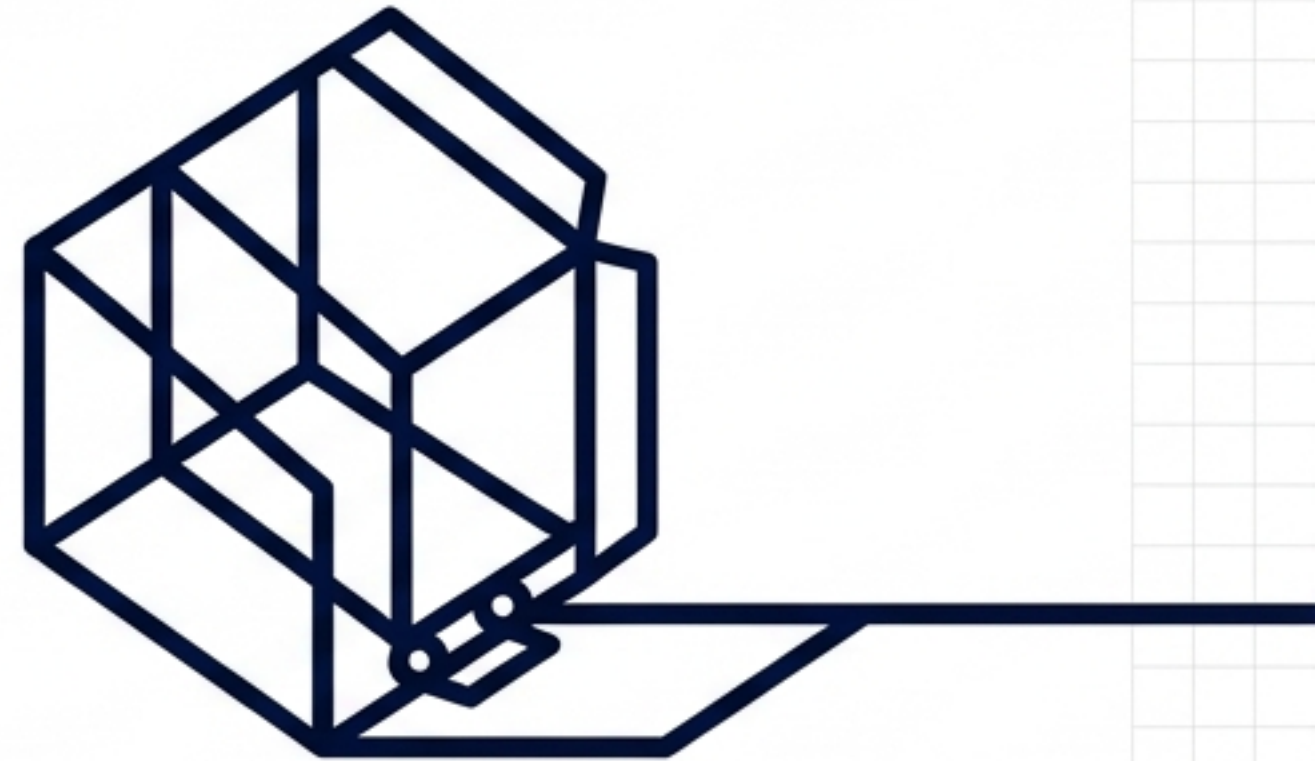


Lab 4.2: Infrastructure Repatriation & Multi- Dimensional Indexing

Designing Decentralized
Discovery using Space-Filling
Curves

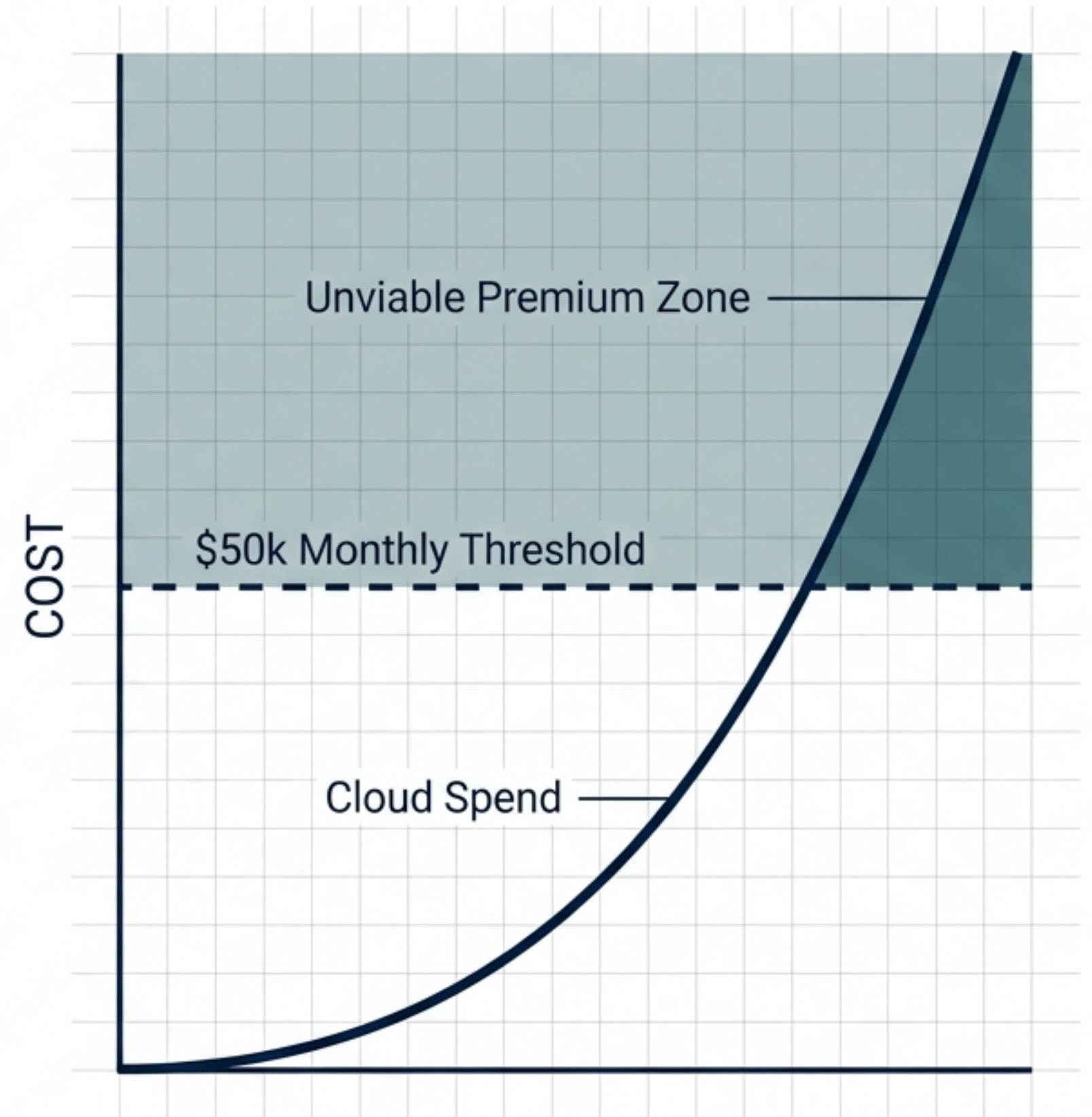
Advanced Systems
Architecture Seminar



The 2026 Inflection Point

Public cloud elasticity has become an operational tax.

- **Cost:** 30–60% savings via dedicated hardware
- **Sovereignty:** Escape shared-tenant performance ceilings
- **The Hurdle:** Losing cloud-native resource discovery

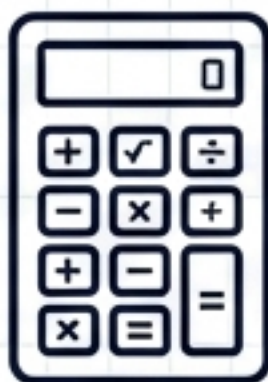


Lab Objectives

By the end of this module, you will:

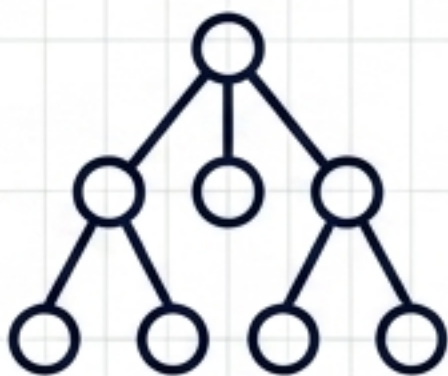
1. Calculate the Total Cost of Ownership (TCO) for repatriated infrastructure.
2. Transition from hierarchical naming (LDAP) to attribute-based discovery.
3. Map multi-dimensional data to a 1D Distributed Hash Table using Hilbert Curves.
4. Architect a private platform that preserves Developer Experience (DX).

Required Lab Tools



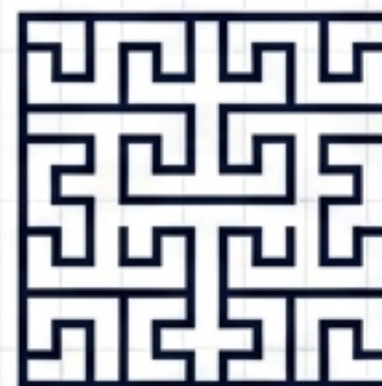
TCO Modeler

Evaluate data egress penalties (15–20%) and GPU hardware premiums (50–70%) against 3-year CAPEX forecasts.



LDAP / DIT Simulator

Analyze search query costs and hierarchical bottlenecks within a localized university structure.



Squid Indexing Framework

Map N-dimensional attributes (e.g., CPU, RAM, Region) to single-dimensional Chord rings.

Step 1: The Financial Architecture

Identify stable workloads. Where does public cloud act as a premium tax rather than a utility?

Public Cloud Centers	Repatriated Advantages
Aggressive data egress fees (15-20%)	Zero data movement costs
Flexibility and on-demand markup	Right-sized, workload-tuned hardware
50–70% markup on GPU/AI hardware	Direct, unimpeded hardware access
Unpredictable monthly billing	3–5 year forecastable CAPEX

Takeaway: At scale, migration is a fiduciary necessity.

Step 2: Pathing vs. Discovery

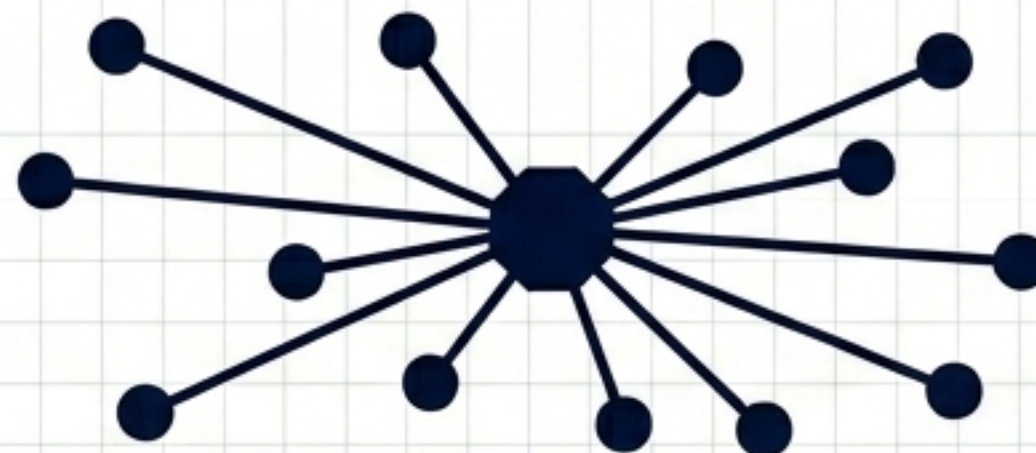
Centralized maps disappear in private infrastructure. Distributed environments rely on what a resource is, not where it resides.

Structured Naming (DNS)



- Unique path or URL string
- Points to a single, specific entity
- Direct 1-to-1 lookup
- Location-dependent

Attribute-Based Naming (Directory)

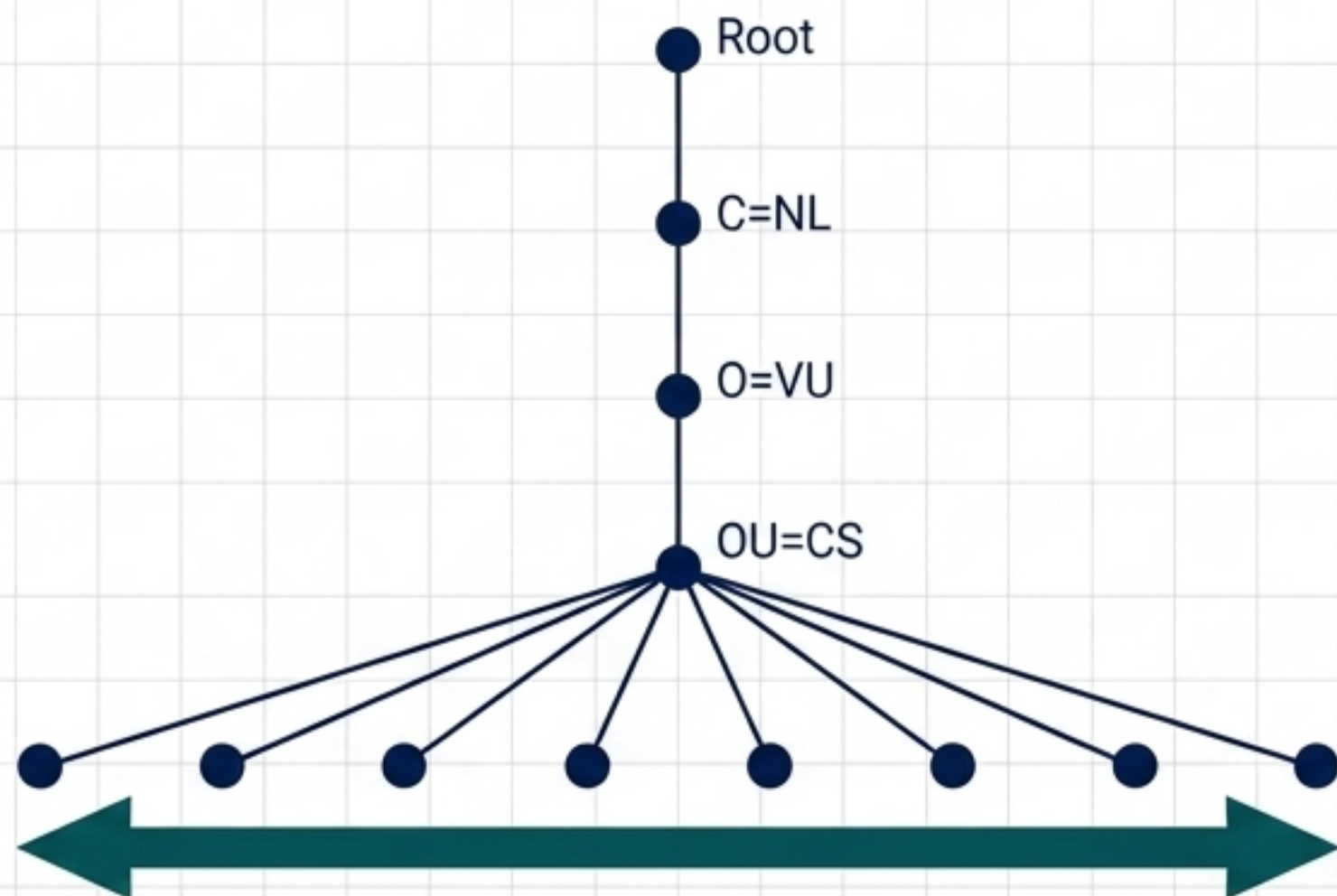


- Collection of (attribute, value) pairs
- Returns multiple matching entities
- Multi-dimensional discovery (1-to-Many)
- Location-independent

Step 3: Hierarchy vs. Decentralization

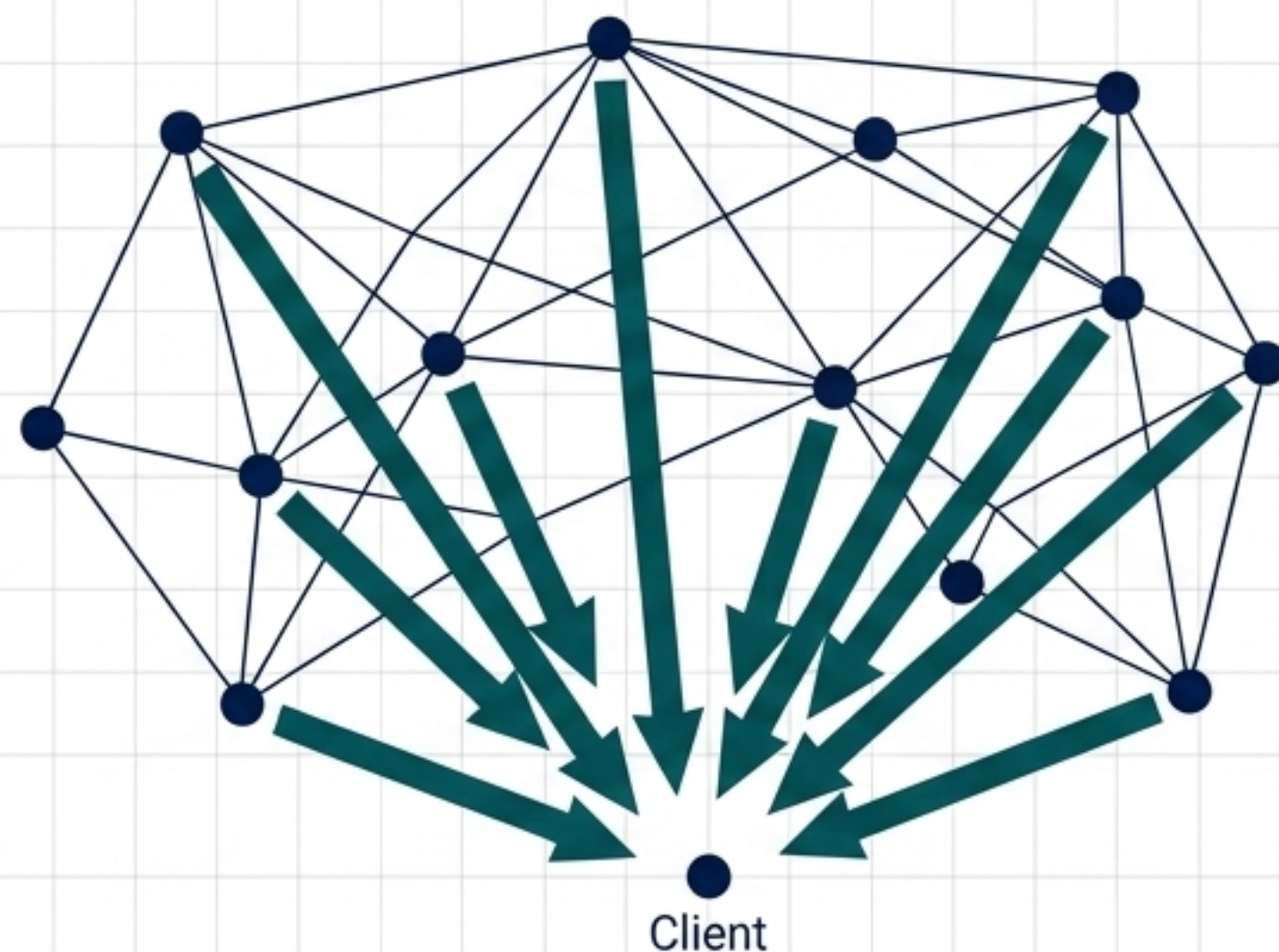
The Hierarchy Problem

The LDAP Bottleneck: Finding specific attributes requires an exhaustive search horizontally across all leaf nodes.



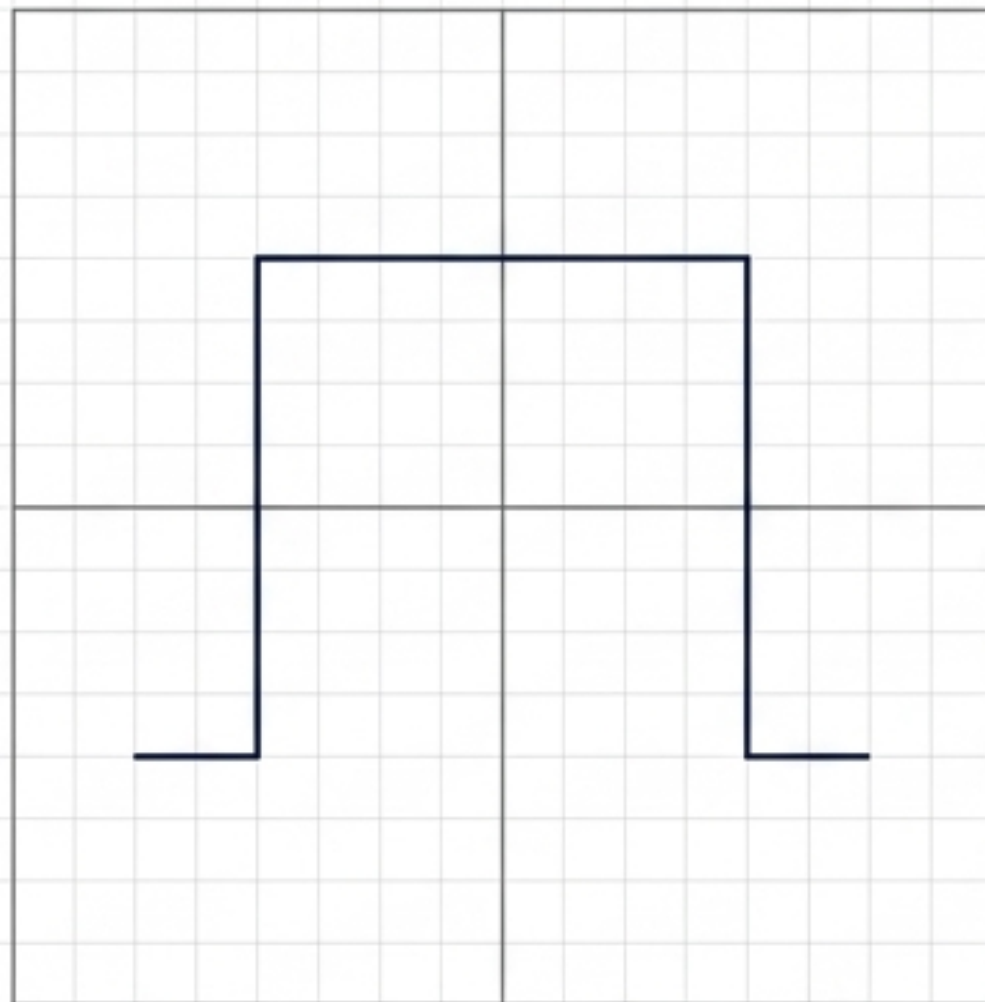
The P2P Bottleneck

The P2P Bottleneck: Simple decentralized indices trigger massive client-side processing loads when filtering partial matches.

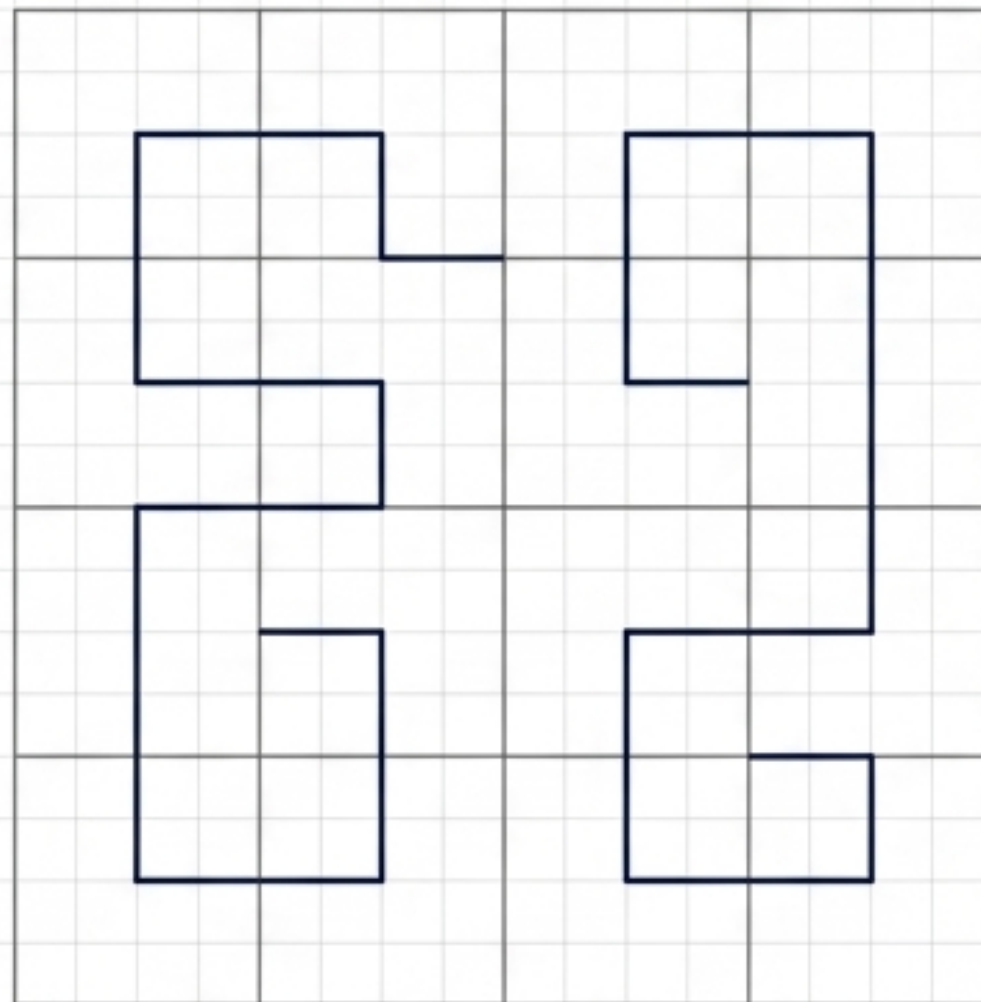


Step 4: The Hilbert Curve Mapping

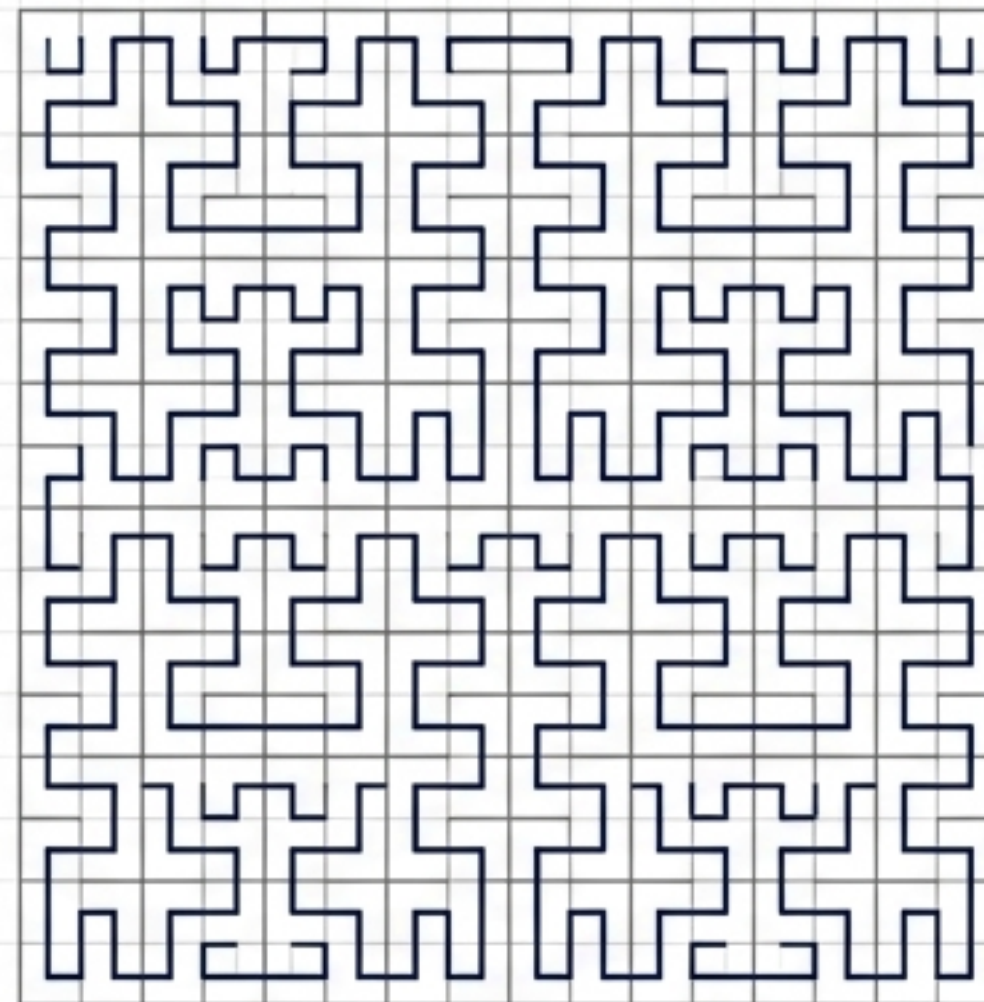
Flatten complex N-dimensional data into a linear index that preserves locality. Items close in multi-dimensional space remain close on the 1D curve.



Order 1 (4 Indices)



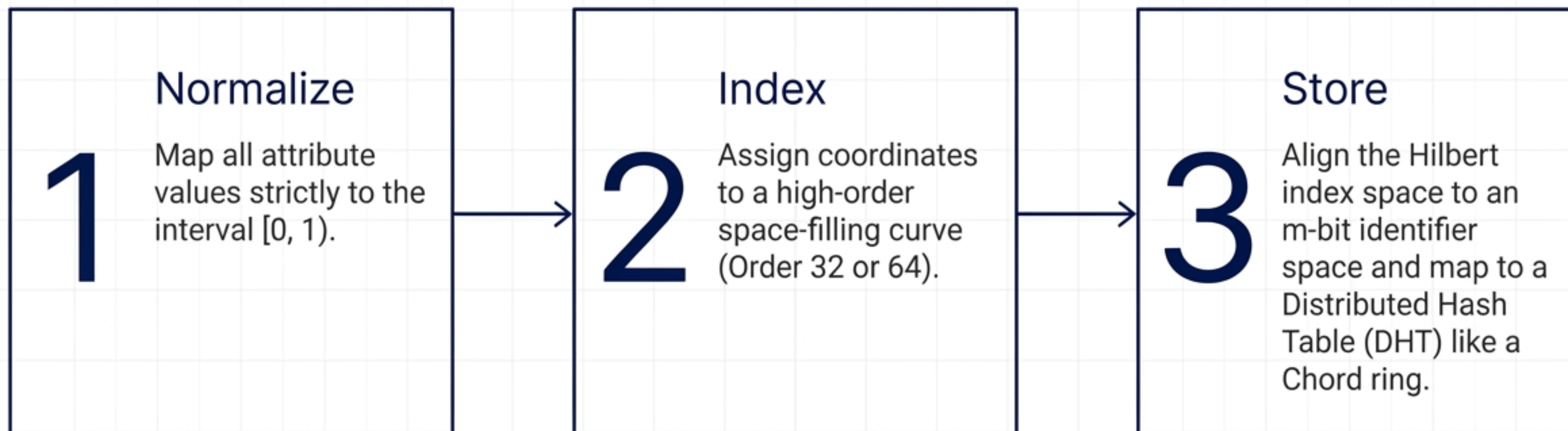
Order 2 (16 Indices)



Order 4 (256 Indices)

Step 5: Implementing the Index

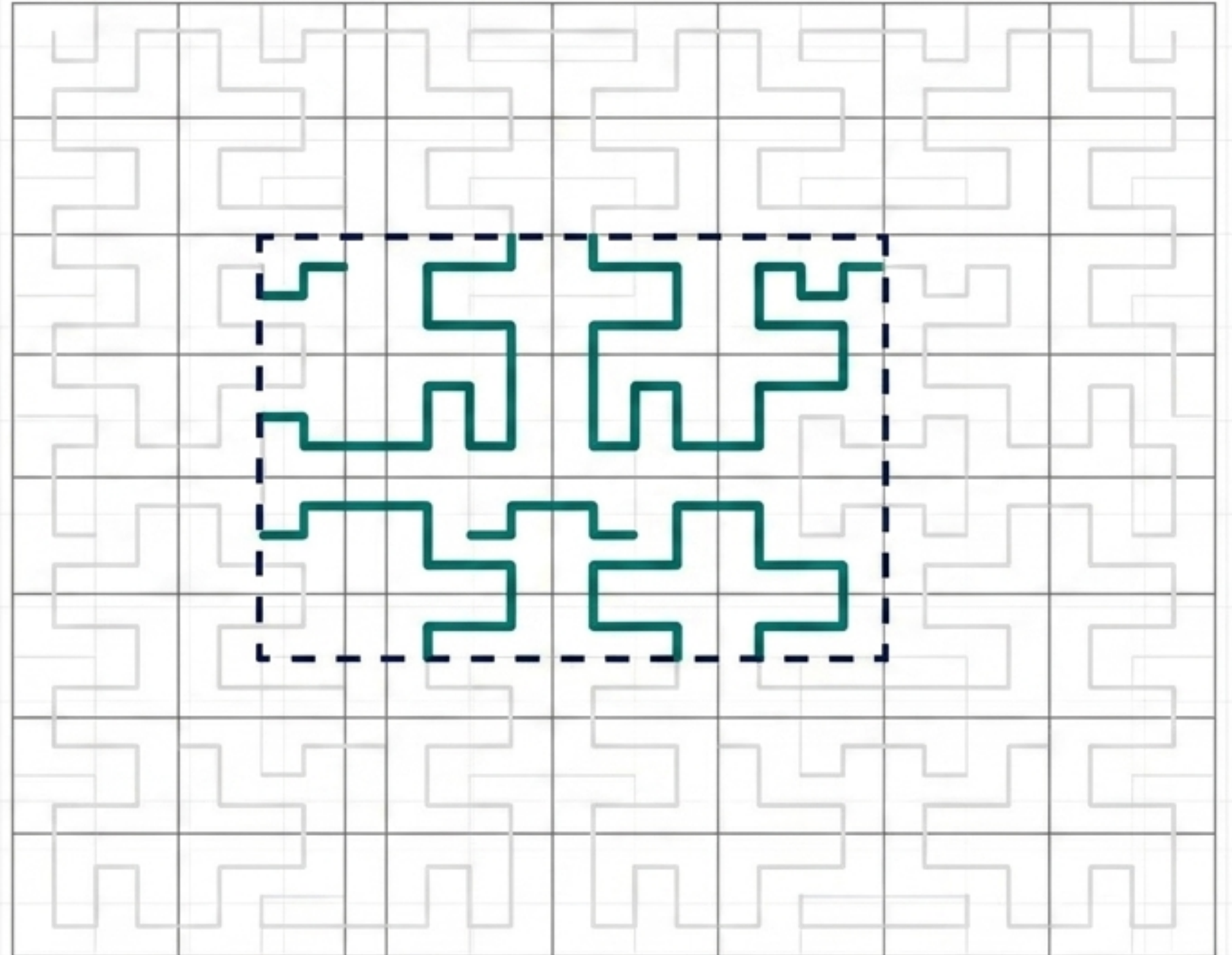
Transitioning from abstract attributes to standard 1D storage systems.



Execution: Handling Range Queries

Queries define a multi-dimensional rectangle. The system calculates which segments of the 1D curve intersect this space.

- Overcomes P2P limitations by supporting range lookups (e.g., Price \$1000–\$2500).
- Queries are routed only to nodes responsible for the intersected curve segments.



Lab Pitfalls: What to Watch For



1. The Locality Caveat

Proximity in N-dimensions does not guarantee curve proximity. Neighbors may lie on separate curve segments, requiring multi-server routing.



2. Developer Experience (DX) Erosion

Repatriation risks destroying engineering velocity by losing built-in cloud orchestration.

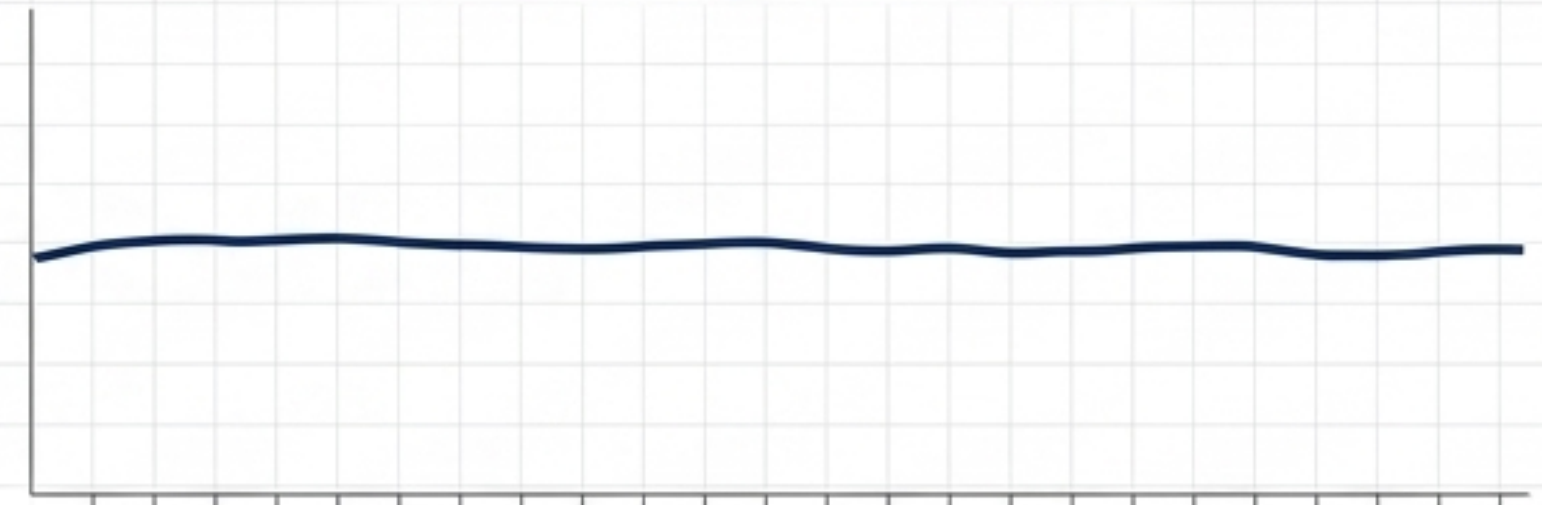
Solution: Layer a BYOC (Bring Your Own Cloud) platform over bare metal to preserve autoscaling and ephemeral environments.

Bonus Challenge: The SWORD System

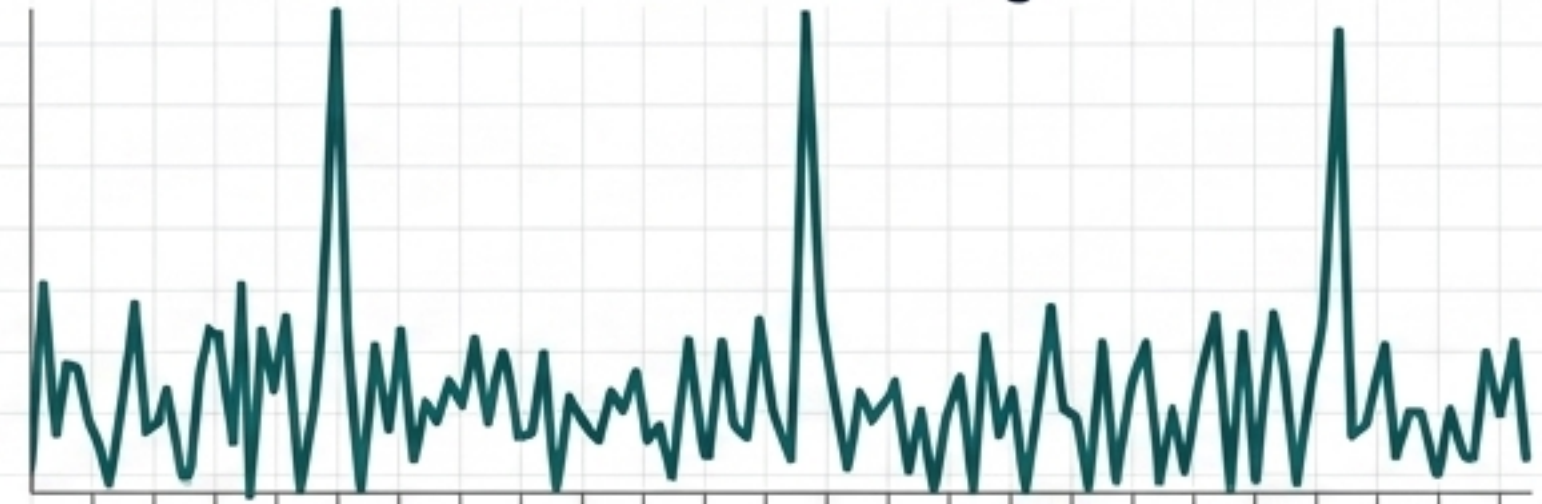
Compare your Hilbert implementation against a bit-partitioning approach.

- **SWORD Mechanics:** Assigns DHT key bits for attributes/values + random bits for uniqueness.
- **The Trade-off:** Supports range queries naturally, but updates must broadcast to multiple servers simultaneously.
- **The Challenge:** Model the “hot server” load-balancing problem caused by popular attribute subranges in SWORD versus the uniform distribution of the Hilbert curve.

Hilbert Uniform Distribution



SWORD Hot Subranges



Server Index

Required Deliverables

Submit the following to the advanced seminar portal:



TCO Analysis: A 3-year projection proving the workload breaches the >30% savings threshold.



Architecture Blueprint: A structural diagram mapping the hierarchical DIT to your proposed Chord ring.



Curve Output: The normalization script for an Order-32 Hilbert mapping of three unique resource attributes.

The Synthesis of Dimensionality

Modern distributed systems achieve infinite scalability by radically reducing dimensions.

By flattening complex descriptions into 1D space, we bridge the gap between cost-effective private hardware and hyper-efficient data discovery.